

# HDOC 15

## Activity Description (Batches 15 & 16):

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even. E.g. 1: Number: 12345, Output: All digits not even. E.g. 2: Number: 24680, Output: All digits are even.
- Check whether all digits are prime. E.g. 1: Number: 2357, Output: All digits are prime. E.g. 2: Number: 2379, Output: All digits are not prime.

Name: \_\_\_\_\_

SAP ID: \_\_\_\_\_

Batch: \_\_\_\_\_

## 1. Algorithm

### 1. Start

2. Declare variables for choice, num, temp, rem, all\_even, and all\_prime.

3. Display the menu options to the user.

4. Read the user's choice.

5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that  $\text{num} > 0$ .

6. Use a switch statement based on the value of choice.

### 7. Case 1 (Check if all digits are even):

- Set `all_even = 1` and `temp = num`.
- Loop while (`temp > 0`):
  - Calculate `rem = temp % 10`.
  - If `rem % 2 != 0`, set `all_even = 0` and break the loop.
  - Update `temp = temp / 10`.
- If `all_even == 1`, display "All digits are even". Else, display "All digits not even".
- Break.

### 8. Case 2 (Check if all digits are prime):

- Set `all_prime = 1` and `temp = num`.
- Loop while (`temp > 0`):
  - Calculate `rem = temp % 10`.

- If rem is not 2, 3, 5, or 7, set all\_prime = 0 and break the loop.

- Update temp = temp / 10.

- If all\_prime == 1, display "All digits are prime". Else, display "All digits are not prime".

- Break.

9. **Default Case:** Display an "Invalid choice" error message.

10. **Stop**

## 2. Test Case Table

| TC ID | Scenario Description                 | Inputs (choice, num)    | Expected Output                                  |
|-------|--------------------------------------|-------------------------|--------------------------------------------------|
| TC-01 | Valid inputs (All even digits)       | Choice: 1<br>Num: 24680 | Output: All digits are even                      |
| TC-02 | Valid inputs (Mixed odd/even)        | Choice: 1<br>Num: 12345 | Output: All digits not even                      |
| TC-03 | Valid inputs (All prime digits)      | Choice: 2<br>Num: 2357  | Output: All digits are prime                     |
| TC-04 | Valid inputs (Mixed prime/composite) | Choice: 2<br>Num: 2379  | Output: All digits are not prime                 |
| TC-05 | Invalid negative number input        | Choice: 1<br>Num: -200  | Error: Please enter a strictly positive integer. |
| TC-06 | Invalid menu choice                  | Choice: 4               | Invalid choice. Please choose 1 or 2.            |

## 3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem;
    int all_even, all_prime;

    // Display Menu
```

```

printf("--- Digit Operations Menu (HDOC 15) ---\n");
printf("1. Check whether all digits are even\n");
printf("2. Check whether all digits are prime\n");
printf("Enter your choice (1 or 2): ");
scanf("%d", &choice);

// Validate Choice and Take Number Input
if (choice == 1 || choice == 2) {
    printf("Enter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer.\n");
        return 1; // Exit program due to invalid input
    }
}

// Switch Case Logic
switch (choice) {
    case 1:
        // Check if all digits are even
        temp = num;
        all_even = 1; // Assume true initially

        while (temp > 0) {
            rem = temp % 10;
            if (rem % 2 != 0) {
                all_even = 0; // Found an odd digit
                break;
            }
            temp /= 10;
        }

        printf("Number: %lld\n", num);
        if (all_even) {
            printf("Output: All digits are even\n");
        } else {
            printf("Output: All digits not even\n");
        }
        break;

    case 2:
        // Check if all digits are prime
        temp = num;
        all_prime = 1; // Assume true initially

        while (temp > 0) {

```

```
        rem = temp % 10;
        // Prime digits are 2, 3, 5, 7
        if (rem != 2 && rem != 3 && rem != 5 && rem != 7) {
            all_prime = 0; // Found a non-prime digit
            break;
        }
        temp /= 10;
    }

    printf("Number: %lld\n", num);
    if (all_prime) {
        printf("Output: All digits are prime\n");
    } else {
        printf("Output: All digits are not prime\n");
    }
    break;

default:
    printf("Invalid choice. Please choose 1 or 2.\n");
    break;
}

return 0;
}
```